

Preliminary descriptive evidence for supervisor discussion. The causal estimation panel is complete and the did package is installed; no CS(2021) coefficient is reported until the att_gt() run completes and its diagnostics are checked.

Three preliminary facts.

- The corrected sample is anchored to the published 513-merger universe. It retains 478 uniquely matched target-spine deals after Cassi's target-side exclusions; 345 lie in the 1993-2010 clean window.
- Among 28,279 pre-deal target inventors in the clean sample, mean annual patenting falls from 0.474 in year -1 to 0.246 in the deal year and 0.113 by year +5. These are descriptive means, not treatment effects.
- The decline is almost entirely an extensive-margin story: the share of the cohort filing any patent falls from 27.5% (year -1) to 4.7% (year +5), while patents per *active* inventor actually rises (1.72 → 2.42 over the same window). See Figure 2.

Corrected sample construction

Stage	Count
Merger-list deals	513
Unique spine matches	490
After target drop flags	478
Clean target-spine deals	345
Clean deals with inventors	338
Clean target inventors	28,279
Status-eligible inventors	27,510

Why the deal count changed

The group-history files cover more mergers than the published sample. The previous reconstruction inadvertently admitted those additional events. The corrected code first anchors each event to deal_map.deal_id, excludes 22 unmatched deals and one ambiguous match, and then applies Cassi's drop flags.

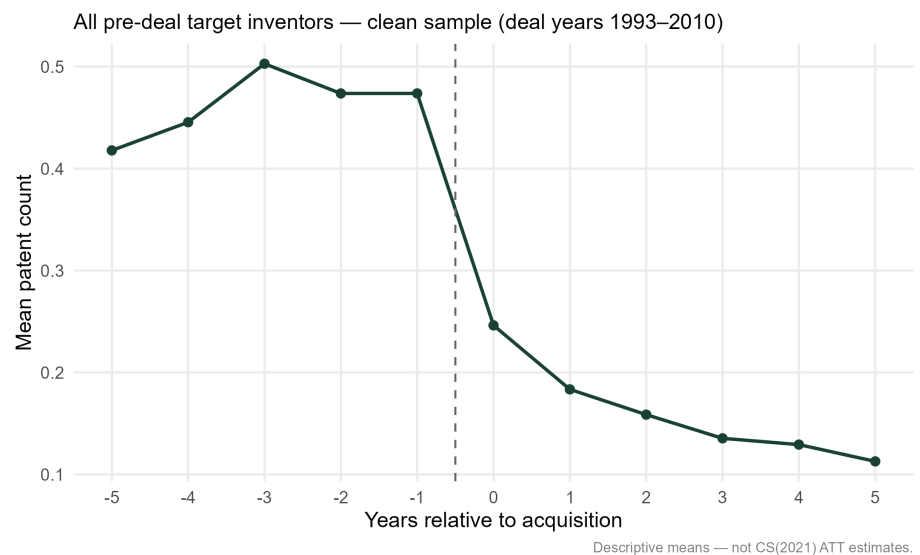


Figure 1. Full pre-deal target cohort. Missing inventor-years are coded as zero patent output. The figure is an event-time profile and contains no comparison group.

Causal estimation status. The balanced CS(2021) panel passes uniqueness and balance checks, and the official did package now installs and loads cleanly. The att_gt() estimation run (not-yet-treated controls, deal-level clustering) is queued next; no CS(2021) coefficient is reported in this memo until that run completes and passes the pre-trend and cohort-support checks.

Full-Cohort Decomposition and Pre-Deal Selection

Both productivity figures use the fixed pre-deal cohort. No displayed productivity trajectory conditions on staying, continued patenting, or another post-deal outcome.

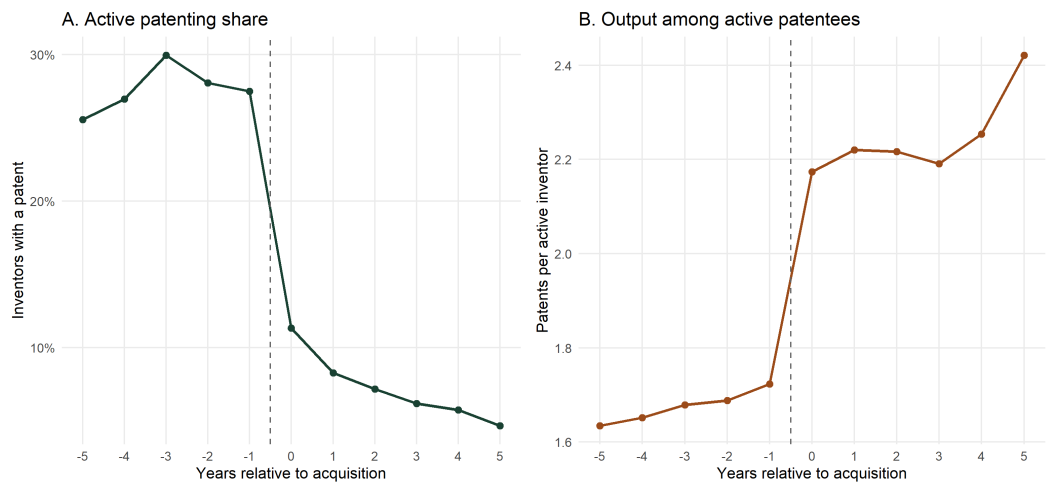


Figure 2. Full-cohort decomposition into active patenting and patents per active inventor. The decline is driven mainly by panel A; panel B describes the changing composition of active patentees and is not a causal intensive-margin estimate.

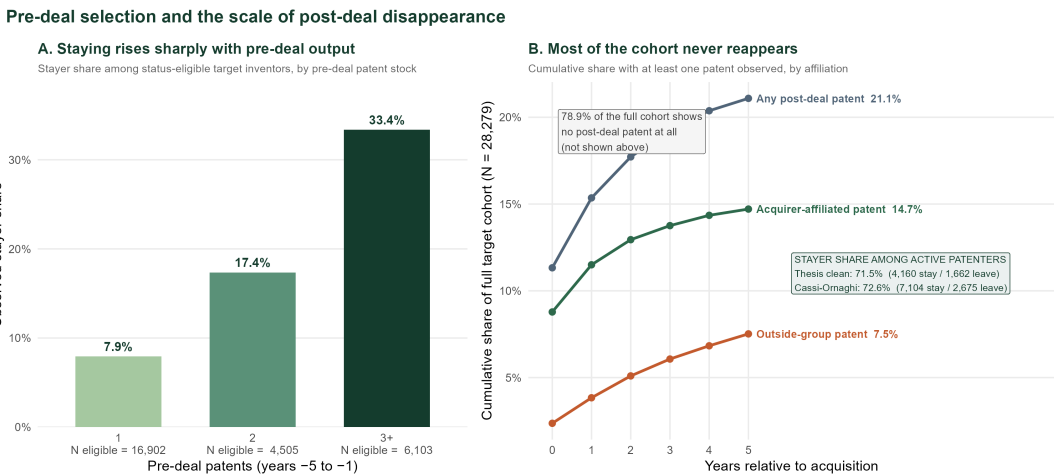


Figure 3. Panel A reports observed staying by pre-deal patent stock. Panel B keeps the full 28,279-inventor denominator and shows cumulative post-deal patent reach. The inset reconciles the apparent retention puzzle: among active post-deal patenters, the thesis stayer share is 71.5% (4,160 stayers; 1,662 leavers), close to Cassi-Ornaghi's 72.6% (7,104; 2,675). Acquirer and outside reach can overlap; all comparisons are descriptive.

Active-patenter benchmark

Source	Stayers	Leavers	Share
Thesis, clean window	4,160	1,662	71.5%
Cassi-Ornaghi	7,104	2,675	72.6%

Only 14.7% of the full cohort is ever acquirer-affiliated because 78.9% never patents again within five years. Conditional on observable post-deal activity, the stay/leave composition closely matches Cassi-Ornaghi.

Interpretation discipline

- Figures 1–2 and Figure 3B retain the full pre-deal cohort.
- Figure 3A is a descriptive selection diagnostic based on pre-deal output.
- T_Ever_Stayed means at least one acquirer-affiliated patent within five years, not continuous retention.
- The CS(2021) estimate for active_patenting must determine whether silence exceeds the counterfactual rate.

Feedback requested from Prof. Cassi.

- Is there an intended mapping for the 22 merger-list deals without an exact group-history match?
- Which Zephyr event corresponds to the ambiguous 2008 acquisition value of 25,000: Curacyte Discovery or Hunter-Fleming?
- When group-history timing differs from the Zephyr legal merger year, should treatment follow organizational integration or legal closing?